Instance-Based vs. Model-Based Learning

Module 6 Study Notes • How Machines Learn

1. Introduction: How do Machines Actually "Learn"?

In previous modules, ML was categorized by the *amount of supervision* and the *production environment*. This categorization, however, is based on the fundamental cognitive approach the algorithm takes to solve a problem.

Just like human students, ML algorithms generally employ one of two strategies when studying data:

- **Memorization:** Learning by rote. Storing the exact examples and recalling them when asked a similar question.
- **Generalization:** Learning the underlying principles or concepts. Deriving a mathematical rule that explains the data, rather than remembering the data itself.

These two strategies translate directly into **Instance-Based Learning** and **Model-Based Learning**.

2. Instance-Based Learning (The "Memorizer")

Also known as "Lazy Learning," this approach involves algorithms that simply memorize the entire training dataset. There is virtually no active "training" phase.

> **How it works:** 1. The algorithm receives the training data and stores it in memory.
> 2. When a new query (a new, unseen data point) is introduced, the algorithm calculates the mathematical distance (similarity) between the new point and all the memorized points.
> 3. It looks at the closest 'neighbors' and assigns the output based on what the majority of those neighbors are.
> *Analogy:* "Tell me who your closest neighbors are, and I will tell you who you are."

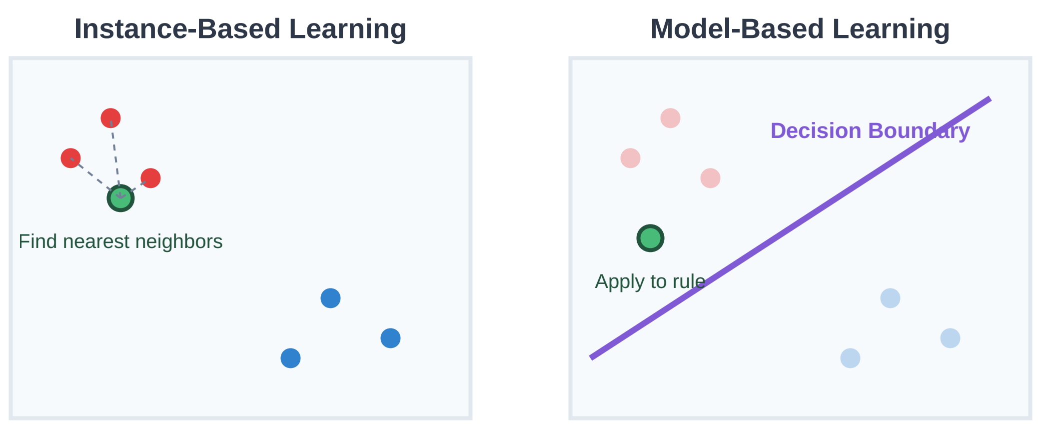
- **Common Algorithm:** K-Nearest Neighbors (KNN).
- **Key Trait:** Training is instant (it just saves data), but predicting is slow (it has to calculate distances to every saved point).

3. Model-Based Learning (The "Generalizer")

This is the more common approach in Machine Learning. Instead of memorizing the data, the algorithm studies the data to discover an underlying pattern or mathematical rule (a "model").

> **How it works:** 1. The algorithm analyzes the training data over many iterations.
> 2. It formulates a mathematical equation or a "Decision Boundary" that cleanly separates or defines the data.
> 3. Once the rule (the model parameters) is established, the algorithm can actually delete or throw away the training data.
> 4. When a new query arrives, it simply plugs the new numbers into the established equation to get an instant prediction.

- **Common Algorithms:** Linear Regression, Logistic Regression, Decision Trees, Neural Networks.
- **Key Trait:** Training is computationally heavy and slow, but predicting is incredibly fast.



4. Key Differences Summary

Feature	Instance-Based Learning	Model-Based Learning
Learning Strategy	Memorizes the data. No generalized rule is created.	Derives a generalized mathematical rule (a model).

Training Phase	Extremely fast (often zero computation, just saving data).	Slow and computationally expensive.
Prediction Phase	Slow. Must compute distance against every saved data point.	Extremely fast. Just plugs numbers into the established equation.
Data Storage Needs	High. The entire training dataset must be kept in production.	Low. Training data can be discarded; only the final model parameters (the rule) are stored.
Handling Queries	Checks similarity with neighbors on the fly for every query.	Applies the predefined decision boundary to the query.